Lab Five-Six – Data Wrangling in R

Tips

- Complete the tasks below. Make sure to start your solutions in on a new line that starts with "**Solution**:".
- Make sure to use the Quarto Cheatsheet. This will make completing and writing up the lab *much* easier.
- Make sure to use "enter" to make your code readable, so it doesn't run off the page. I switched the Quarto document to automatically render to .pdf, so you can see what you're submitting by default. You can switch back for the highlighting by moving the html block above the pdf block in the YAML header.
- Use the pipe operator (`|>`). It makes the data wrangling tasks much easier to write.
- Use `head(...)` or `glimpse(...)` often to ensure your code did what you wanted it to and that you haven't accidentally deleted their entire dataset with a bad filter.
- Don't forget that you can copy-and-paste code when you're doing something similar as we did in class!

Timeline

- Week of 2/16: Aim to complete the data wrangling sections
- Week of 2/23: Aim to complete the data summaries sections

1 Biomedical Science Data

Kitsberg et al. (2025) collect data on the Human Cytomegalovirus (HCMV). HCMV is a member of the herpesvirus family – the same family that includes: chickenpox, mononucleosis (mono), and cold sores. Prevalence is high with a worldwide seroprevalence of 24% to 100% (Gupta and Shorman 2025). The virus persists in a dormant state for the entire life of those infected. Primary cytomegalovirus infection is often asymptomatic but can present with flu-like symptoms. The infection can reactivate due to stress, aging, or a weakened immune system.

The researchers obtained blood samples from healthy donors, males and females, aged 25-45. They infected the cells of primary (human donor's blood) or THP1 (purchased cell lines) monocytes ($n = 87$ and $n = 112$, respectively) and macrophages ($n = 93$ and $n = 170$, respectively) and counted the number of viral genomes in the nuclei.

Note that the original donor of the THP-1 cell line was a 1-year-old Japanese boy. He was treated for Acute Monocytic Leukemia at Tohoku Hospital Pediatrics (THP-1; the first cell line there). Because they are malignant cancer cells, they can divide indefinitely, breaking the usual cell managment that usually reigns in rogue cells. Therefore, these cells can divide indefinitely in a lab environment. THP-1 cells, and those like them, serve as such a reliable, uniform model for medical research and improve reproducability.

Preliminary Question: Are THP-1 cells a good stand-in for human cells?

Research Question: Is the virus more prevalent in the nucleus of monocytes or macrophages, explaining why HCMV stays dormant in monocytes but goes active in macrophages?

1.1 Data Wrangling

1.1.1 Part A

Go to the website and download the source data https://www.nature.com/articles/s41467-025-68063-y#Sec30.

Describe the process – do not leave out any steps – for how you can load the data for Figures 3b and 3d from the manuscript into `R`.

Hint: You can use packages like `readxl` (Wickham and Bryan 2025) but you will find it substantially easier to open the `.xls` file using your computer's spreadsheet software and create a new `.csv` file with the data you need.

Solution

First, I downloaded the data from the website and opened in it excel. It came in two sections, and each had two columns of data. I copied all the data and pasted it into a new excel file. Then, I moved the data to all be in the same two columns. I saved this file to the lab folder. After that, I read the data using read_csv, and looked at it with head() to get an idea of what the columns were named.

```
data.kitsberg = read_csv("Figure3bData.csv")
head(data.kitsberg)
```

```
# A tibble: 6 x 2
  sample                  `viruses within nucleus`
  <chr>                                      <dbl>
1 primary_mono_uninfected                        0
2 primary_mono_uninfected                        0
3 primary_mono_uninfected                        0
4 primary_mono_uninfected                        0
5 primary_mono_uninfected                        0
6 primary_mono_uninfected                        0
```

1.1.2 Part B

The data need to be cleaned. Note that the sample type (Healthy Donor or THP-1), cell type (Monocytes or Macrophages), and treatment (12 Hours Post-Infection or Uninfected) are all recorded in one column `sample`, separated with underscores. Use `separate(...)` from the `tidyverse` package to separate the `sample` column into the `SampleType`, `CellType`, `Treatment` columns. Save the object as `dat.kitsberg.cleaned`.

Hint: Use `?separate(...)` to see how it works.

Solution

In this code, I use separate to split the column into three by underscore.

```
dat.kitsberg.cleaned = data.kitsberg |>
  separate(col = sample, into= c("SampleType", "CellType", "Treatment"), sep = '_' )
```

1.1.3 Part C

The researchers are only interested in observations in the treatment group (12hpi). Remove uninfected samples from the data. Update `dat.kitsberg.cleaned` with the result.

Hint: I would use the `filter(...)` function.

Solution

Here, I use filter to remove all rows that do not have treatment of 12hpi.

```
dat.kitsberg.cleaned = dat.kitsberg.cleaned |> filter(Treatment=="12hpi")
```

1.1.4 Part D

When `CellType` is "MDM", it is a Monocyte-Derived Macrophages. That is, the cell type should be macrophages which is coded as "mac". Update `dat.kitsberg.cleaned` with the result.

Hint: I would use the `mutate(...)` function. You will likely find `if_else(...)` or `case_when(...)` helpful.

Solution

In this problem, I used mutate and case_when to look through all entries in column CellType and change entries that were "MDM" to "mac". Otherwise, they remained the same.

```
dat.kitsberg.cleaned = dat.kitsberg.cleaned |>
  select(SampleType, CellType, Treatment, 'viruses within nucleus') |>
  mutate(
```

```
    CellType = case_when(
      CellType == "MDM" ~ "mac",
      .default = CellType
    )
  )
```

1.2 Part E

The researchers are only interested in monocytes and macrophages. Remove any other cell types. Update `dat.kitsberg.cleaned` with the result.

Hint: I would use the `filter(...)` function.

Solution

In this code, I filter out all values that do not have a CellType value of "mac" or "mono."

```
dat.kitsberg.cleaned = dat.kitsberg.cleaned |>
  filter(CellType =="mac" | CellType=="mono")
```

1.3 Part F

Clean up the data entry. Replace the shorthand the researchers used with the neat sample type (Healthy Donor or THP-1), cell type (Monocytes or Macrophages), and treatment (12 Hours Post-Infection or Uninfected). Update `dat.kitsberg.cleaned` with the result.

Solution

In this code, I use the mutate() and case_when() functions like I did previously to change shorthand to the neat sample type described in the question.

```
dat.kitsberg.cleaned = dat.kitsberg.cleaned |>
  mutate( # section to change CellType into the shorthand
    CellType = case_when(
      CellType == "mac" ~ "Macrophage",
      CellType == "mono" ~ "Monoctye",
      .default = CellType
    )
  ) |>
  mutate( #section to change Treatment into the shorthand
    Treatment = case_when(
      Treatment == "12hpi" ~ "12 Hours Post-Infection",
      .default = Treatment
    )
  ) |>
  mutate( #changing SampleType into the shorthand
    SampleType = case_when(
      SampleType == "primary" ~ "Healthy Donor",
      SampleType == "THP1" ~ "THP-1",
      .default = SampleType
    )
  )
```

Hint: I would use the `mutate(...)` function. You will likely find `if_else(...)` or `case_when(...)` helpful.

1.4 Part G

In an `R` code block, combine your answers into one code block that completes the full cleaning process. Save the object as `dat.kitsberg.cleaned`.

Solution

In this code, I combine all my previous code and pipe it together. It produces the same result as the individual code.

```
dat.kitsberg.cleaned = data.kitsberg |>
  # the below code is the code i used earlier
  # it separates the column, and then filters the data
  # so only the desired columns are chosen
  separate(col = sample, into= c("SampleType", "CellType", "Treatment"), sep = '_' ) |>
  filter(Treatment=="12hpi") |>
  select(SampleType, CellType, Treatment, 'viruses within nucleus') |>
  mutate(
    CellType = case_when(
      CellType == "MDM" ~ "mac",
      .default = CellType
    )
  ) |>
  filter(CellType =="mac" | CellType=="mono") |>
  mutate(
    CellType = case_when(
      CellType == "mac" ~ "Macrophage",
      CellType == "mono" ~ "Monocyte",
      .default = CellType
    )
  ) |>
  mutate(
    Treatment = case_when(
      Treatment == "12hpi" ~ "12 Hours Post-Infection",
      .default = Treatment
    )
  ) |>
  mutate(
    SampleType = case_when(
      SampleType == "primary" ~ "Healthy Donor",
      SampleType == "THP1" ~ "THP-1",
      .default = SampleType
    )
  )


dat.kitsberg.cleaned <- dat.kitsberg.cleaned |>
  mutate(CellType2 = factor(CellType,
                            levels=c('Monocyte', 'Macrophage'))
  )
```

Note: You will use this in the next section

1.5 Data Summaries

1.5.1 Part A

Recreate Figure 3b as depicted in the paper (see https://www.nature.com/articles/s41467-025-68063-y#Fig3). You can ignore the significance comparison because we haven't done that yet.

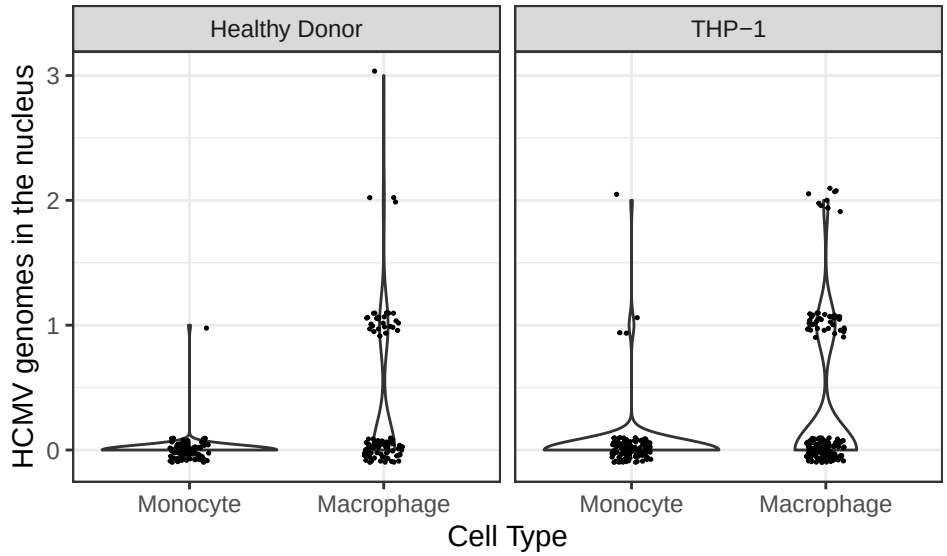
Hint: I would use `geom_violin(...)` for Figure 3b I also expect your plots to look nicer – better labels, better scaling, etc. Don't forget to use the proper formatting using the syntax in the Quarto cheatsheet.

Solution This figure tells us that there is a noticable difference in HCMV genomes in the nucleus between monocytes and macrophages in a healthy patient, but that difference is not visible in the monocytes and macrophages 12 hours after infection.

```
ggplot(data = dat.kitsberg.cleaned) +
  geom_violin(aes(x=CellType2, y = `viruses within nucleus`))+
```

```
  geom_jitter(aes(x=CellType2, y = `viruses within nucleus`), height = .1, width = .1, size=0.25) +
  theme_bw() +
  labs(x = "Cell Type", y = "HCMV genomes in the nucleus")+
  facet_wrap(~SampleType)
```

Figure 1: Viruses within nucleus by Cell Type and Sample Type.



1.5.2 Part B

Compute a table containing relevant numerical summaries for Figure 3b.

Hint: I would use `group_by(...)` and `summarize(...)` to complete this. Don't forget to use the proper formatting using `knitr::kable(...)` as described in the Quarto Cheatsheet.

Solution

Here is the table I made containing relevant summary statistics.

```
dat.kitsberg.cleaned |>
    group_by(CellType2, SampleType) |>
    summarize(
        Mean = mean(`viruses within nucleus`),
        SD = sd(`viruses within nucleus`),
        Median = median(`viruses within nucleus`),
        IQR = IQR(`viruses within nucleus`),
    )|>
    knitr::kable(digits = 2)
```

Table 1: Viruses within nucleus by Cell Type and Sample Type.

CellType2	SampleType	Mean	SD	Median	IQR
Monocyte	Healthy Donor	0.01	0.11	0	0
Monocyte	THP-1	0.04	0.25	0	0
Macrophage	Healthy Donor	0.37	0.60	0	1
Macrophage	THP-1	0.33	0.57	0	1

1.5.3 Part C

Recreate Figure 3d as depicted in the paper (see https://www.nature.com/articles/s41467-025-68063-y#Fig3). You can ignore the significance comparison because we haven't done that yet.

Hint: I would use `geom_jitter(...)` and `geom_crossbar(...)` for Figure 3d. I also expect your plots to look nicer – better labels, better scaling, etc. Don't forget to use the proper formatting using the syntax in the Quarto cheatsheet.

1.5.4 Part D

Compute a table containing relevant numerical summaries for Figure 3d.

Hint: I would use `group_by(...)` and `summarize(...)` to complete this. Don't forget to use the proper formatting using `knitr::kable(...)` as described in the Quarto Cheatsheet.

2 Ship Valuation

Figure 2: This is a ship-shipping ship shipping shipping ships.

Bal et al. (2025) aim to build a model for predicting the value of Supramax and Ultramax ships using both current and past market data. Their goal was to build a model that only uses information anyone can easily look up.

The researchers collected 17 years of real ship sale prices (2005–2022), covering $n = 1819$ Supramax and Ultramax ship sales.

Research Question: Can we predict Supramax and Ultramax ship prices?

2.1 Data Wrangling

2.1.1 Part A

I downloaded the data from linked in the article (https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0319073). I have included the data in this lab folder as `ShipValuation-Bal25.csv`. Load the data into `R`.

Solution

In this code I read the data into R.

```
ship_data=read_csv("ShipValuation-Bal25.csv")
```

2.1.2 Part B

The price of each Supramax and Ultramax ship is reported on the $\log_{10}$ scale. Create a new column called `price` that contains the price in millions of dollars.

$$\text{Price} = \frac{10^{\text{SPLog}}}{1000000}$$

Save the object as `dat.bal.cleaned`.

Hint: I would use the `mutate(...)` function.

Solution

In this code I use the given equation to create a new column with the modified data.

```
dat.bal.cleaned = ship_data |>
  mutate(
    price = 10^SPLog/1000000
  )
```

2.1.3 Part C

The authors recorded the country that each Supramax or Ultramax ship was built in in the columns `Japan`, `China, People's Republic Of`, `Philippines`, `Korea, South`, `Others Country Build`. These columns are marked `1` to indicate that is the country of origin and `0` otherwise. Create a new column called `OriginCountry` that contains a clean labels for the country in which the ship was built. Update `dat.bal.cleaned` with the result.

Hint: I would use the `mutate(...)` function. You will likely find `if_else(...)` or `case_when(...)` helpful.

Solution

In this code, I use mutate to create a new column that has the name of the country based on which row in that column had 1 under it's respective country.

```
dat.bal.cleaned = dat.bal.cleaned |>
  mutate(
    OriginCountry = case_when(
      Japan == 1 ~ "Japan",
      `China, People's Republic Of` == 1 ~ "China",
      Philippines == 1 ~ "Philippines",
      `Korea, South`== 1 ~ "South Korea",
      `Others Country Build` == 1 ~ "Other",
    )
  )
```

2.1.4 Part D

The authors recorded each Supramax and Ultramax ship's main engine manufacturer in the columns `Caterpillar`, `MAN-B&W`, `Wartsila`, `Mitsubishi`. These columns are marked `1` to indicate that is the manufacturer and `0` otherwise. Create a new column called `Manufacturer` that contains a clean labels for the manufacturer. Update `dat.bal.cleaned` with the result.

Hint: Follow the logic of the previous question.

In this code, I use the same code as previously but update it to fit the new variables.

```
dat.bal.cleaned = dat.bal.cleaned |>
  mutate(
    Manufacturer = case_when(
      Caterpillar == 1 ~ "Caterpillar",
      `MAN-B&W` == 1 ~ "MAN-B&W",
      Wartsila == 1 ~ "Wartsila",
      Mitsubishi== 1 ~ "Mitsubishi",
```

```
    )
  )
```

2.1.5 Part E

The researchers recorded whether each Supramax and Ultramax ship is currently covered by a Protection and Indemnity (P&I) Club. This can be an indicator that a ship meets the minimum seaworthiness and safety requirements of the insurer.

The authors recorded P&I clubs in the columns `Japan P&I Club`, `Gard`, `West of England`, `North of England`, `UK P&I`, `Britannia P&I`, `London P&I Club`, `Skuld`, `Steamship Mutual`, `Others`, `Unknown`. These columns are marked `1` to indicate that the ship is insured by that P&I club and `0` otherwise. Create a new column called `PI.club` that contains a clean labels for the P&I club. Update `dat.bal.cleaned` with the result.

Hint: Follow the logic of the previous questions.

Solution

In this code, I use the same code as previously but update it to fit the new variables.

```
dat.bal.cleaned = dat.bal.cleaned |>
  mutate(
    PI.club = case_when(
      `Japan P&I Club` == 1 ~ "Japan P&I Club",
      Gard == 1 ~ "Gard",
      `West of England` == 1 ~ "West of England",
      `North of England`== 1 ~ "North of England",
      `UK P&I` == 1 ~ "UK P&I",
      `Britannia P&I` == 1 ~ "Britannia P&I",
      `London P&I Club` == 1 ~ "London P&I Club",
      Skuld == 1 ~ "Skuld",
      `Steamship Mutual` == 1 ~ "Steamship Mutual",
      Others == 1 ~ "Others",
      Unknown == 1 ~ "Unknown",

    )
  )
```

2.1.6 Part F

The researchers recorded whether each Supramax and Ultramax ship is classed by a society that requires a ship to be safe, structurally sound, and built/maintained to specific engineering standards.

The authors recorded class society in the columns `Nippon Kaiji Kyokai (IACS)`, `Lloyd's Register`, `Bureau Veritas (IACS)`, `Det Norske Veritas (IACS)`, `American Bureau of Shipping (IACS)`, `Others Class`, `With Two Classes`, `Disclassed`. These columns are marked `1` to indicate that the ship is classed by that soceity and `0` otherwise. Create a new column called `Class` that contains a clean labels for the class. Update `dat.bal.cleaned` with the result.

Hint: Follow the logic of the previous questions.

Solution

In this code, I use the same code as previously but update it to fit the new variables.

```
dat.bal.cleaned = dat.bal.cleaned |>
  mutate(
    Class = case_when(
      `Nippon Kaiji Kyokai (IACS)` == 1 ~ "Nippon Kaiji Kyokai (IACS)",
      `Lloyd's Register` == 1 ~ "Lloyd's Register",
      `Bureau Veritas (IACS)` == 1 ~ "Bureau Veritas (IACS)",
      `Det Norske Veritas (IACS)`== 1 ~ "Det Norske Veritas (IACS)",
```

```
      `American Bureau of Shipping (IACS)` == 1 ~ "American Bureau of Shipping (IACS)",
      `Others Class` == 1 ~ "Others Class",
      `With Two Classes` == 1 ~ "With Two Classes",
      Disclassed == 1 ~ "Disclassed",
    )
  )
```

2.1.7 Part G

The researchers recorded the flag state of each Supramax and Ultramax ship. Each flag state enforces its own set of safety, environmental, and crew welfare rules and regulations. Ships under certain flag states may be more or less valuable depending on how much the flag affects the buyer's trust in the condition of the boat.

The authors recorded flag states in the columns `Panama`, `Marshall Islands`, `Liberia`, `Malta`, `Hong Kong`, `China`, `Singapore`, `Others Flag`, `With two flags`. These columns are marked `1` to indicate that the ship has that flag state and `0` otherwise. Create a new column called `FlagState` that contains a clean labels for the flag state. Update `dat.bal.cleaned` with the result.

Hint: Follow the logic of the previous questions.

Solution

In this code, I use the same code as previously but update it to fit the new variables.

```
dat.bal.cleaned = dat.bal.cleaned |>
  mutate(
    FlagState = case_when(
      Panama == 1 ~ "Panama",
      `Marshall Islands` == 1 ~ "Marshall Islands",
      Liberia == 1 ~ "Liberia",
      Malta== 1 ~ "Malta",
      `Hong Kong, China` == 1 ~ "Hong Kong, China",
      Singapore == 1 ~ "Singapore",
      `Others Flag` == 1 ~ "Others Flag",
      `With two flags` == 1 ~ "With two flags",
    )
  )
```

2.2 Part H

In an `R` code block, combine your answers into one code block that completes the full cleaning process. Save the object as `dat.bal.cleaned`.

Solution

I combined all my previous code using pipes. I ran it, and it produced the same result as the code individually.

```
dat.bal.cleaned = ship_data[1:1819, ] |>
  mutate(
    price = 10^SPLog/1000000
  ) |>
    mutate(
    OriginCountry = case_when(
      Japan == 1 ~ "Japan",
      `China, People's Republic Of` == 1 ~ "China",
      Philippines == 1 ~ "Philippines",
      `Korea, South`== 1 ~ "South Korea",
      `Others Country Build` == 1 ~ "Other",
    )
  ) |>
```

```r
    mutate(
    Manufacturer = case_when(
      Caterpillar == 1 ~ "Caterpillar",
      `MAN-B&W` == 1 ~ "MAN-B&W",
      Wartsila == 1 ~ "Wartsila",
      Mitsubishi== 1 ~ "Mitsubishi",
    )
  ) |>
  mutate(
    PI.club = case_when(
      `Japan P&I Club` == 1 ~ "Japan P&I Club",
      Gard == 1 ~ "Gard",
      `West of England` == 1 ~ "West of England",
      `North of England`== 1 ~ "North of England",
      `UK P&I` == 1 ~ "UK P&I",
      `Britannia P&I` == 1 ~ "Britannia P&I",
      `London P&I Club` == 1 ~ "London P&I Club",
      Skuld == 1 ~ "Skuld",
      `Steamship Mutual` == 1 ~ "Steamship Mutual",
      Others == 1 ~ "Others",
      Unknown == 1 ~ "Unknown",

    )
  ) |>
  mutate(
    Class = case_when(
      `Nippon Kaiji Kyokai (IACS)` == 1 ~ "Nippon Kaiji Kyokai (IACS)",
      `Lloyd's Register` == 1 ~ "Lloyd's Register",
      `Bureau Veritas (IACS)` == 1 ~ "Bureau Veritas (IACS)",
      `Det Norske Veritas (IACS)`== 1 ~ "Det Norske Veritas (IACS)",
      `American Bureau of Shipping (IACS)` == 1 ~ "American Bureau of Shipping (IACS)",
      `Others Class` == 1 ~ "Others Class",
      `With Two Classes` == 1 ~ "With Two Classes",
      Disclassed == 1 ~ "Disclassed",
    )
  ) |>
  mutate(
    FlagState = case_when(
      Panama == 1 ~ "Panama",
      `Marshall Islands` == 1 ~ "Marshall Islands",
      Liberia == 1 ~ "Liberia",
      Malta== 1 ~ "Malta",
      `Hong Kong, China` == 1 ~ "Hong Kong, China",
      Singapore == 1 ~ "Singapore",
      `Others Flag` == 1 ~ "Others Flag",
      `With two flags` == 1 ~ "With two flags",
    )
  )
```

Note: You will use this in the next section

2.3 Data Summaries

First, I provide a short description of the other variables in their data. Note that the size ordering of types of ships discussed below is:

$$\text{Handysize} < \text{Supramax} < \text{Ultramax} < \text{Panamax} < \text{Capesize}$$

- `DWT`: The deadweight tonnage, which is the maximum weight each Supramax and Ultramax ship can safely carry in metric tons. This gives a measure of how much cargo, fuel, fresh water, ballast water, provisions, and crew a ship can handle.
- `Yas`: The age in years. New Supramax and Ultramax ships were recorded as having age 1.
- `Enginges RPM`: Revolutions per minute of each Supramax and Ultramax ship's main propulsion engine. Generally speaking, lower-RPM engines are more expensive because they are more fuel efficient.
- `BACI`: The Baltic Capesize Index. This describes freight rates for Capesize dry bulk ships, which reflects supply and demand. When BACI is high, Capesize ships are more profitable in shipping, which could positively affect Supramax and Ultramax ship values as demand for dry bulk commodities might be high and large shipments could be split into several Supramax and Ultramax shipments.
  Note: This reports the current BACI. The data also contain the BACI form 1, 2, and 3 months before the sale (`BACI-1`, `BACI-2`, `BACI-3`).
- `BPNI`: The Baltic Panamax Index. This describes the cost to ship dry bulk commodities (e.g., coal or grain) on Panamax class ships. When BPNI is high, we expect Panamax ship prices to go up. Supramax and Ultramax ship values tend to follow this trend as charterers switch to these smaller vessels when Panamax rates become too expensive.
  Note: This reports the current BPNI. The data also contain the BPNI form 1, 2, and 3 months before the sale (`BPNI-1`, `BPNI-2`, `BPNI-3`).
- `BSIS`: The Baltic Supramax Index. This describes freight rates for Supramax vessels, which reflects supply and demand. When BSIS is high, Supramax and Ultramax prices are usually higher because they are more profitable in shipping.
  Note: This reports the current BSIS. The data also contain the BSIS form 1, 2, and 3 months before the sale (`BSIS-1`, `BSIS-2`, `BSIS-3`).
- `BHSI`: The Baltic Handysize Index. This describes freight rates for handysize vessels. When BHSI is high, Supramax and Ultramax ship prices are usually higher because the prices move together.
  Note: This reports the current BHSI. The data also contain the BHSI form 1, 2, and 3 months before the sale (`BHSI-1`, `BHSI-2`, `BHSI-3`).
- `LIBOR`: The London InterBank Offered Rate. This describes the average reported interest rate at which banks would lend US dollars for 3 months.

2.3.1 Part A

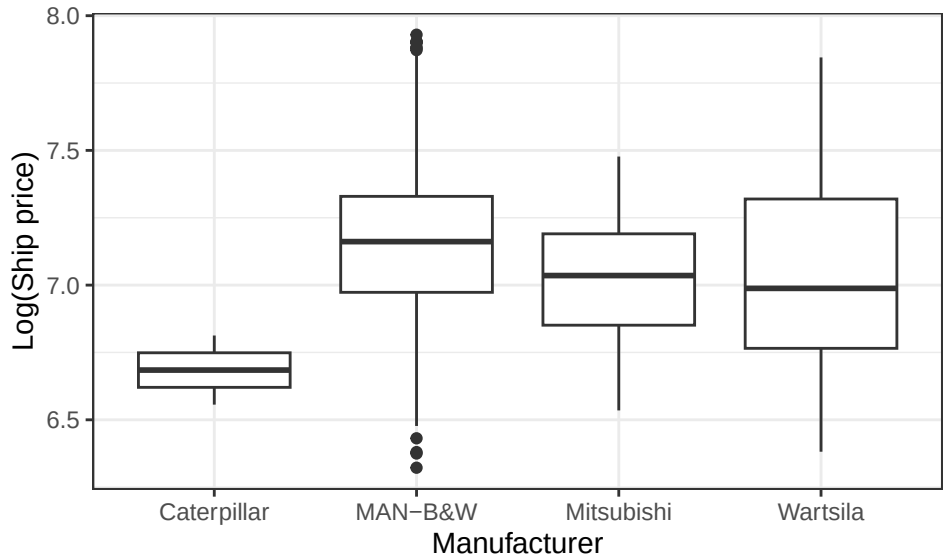
Choose a categorical variable. Create a plot and corresponding numerical summary to assess whether or how ship prices vary across the levels of the selected categorical variable.

Solution

I am choosing the manufacturer of the ship as my categorical variable. Here is my grouped boxplot.

```r
ggplot(data = dat.bal.cleaned) +
  geom_boxplot(aes(x = Manufacturer, y = SPLog)) +
  theme_bw() +
  labs(x = "Manufacturer", y = "Log(Ship price)")
```

Figure 3: Log of ship price by manufacturer



In this plot, we notice that MAN-B&W, Mitsubishi, and Wartsila all have roughly similar median ship prices, with Caterpillar having notably cheaper prices than the other three. Interestingly, MAN-B&W has the widest range of ship prices, as well as the most expensive and cheapest ships, despite having the highest median price. It looks like Wartsila has the largest IQR, Caterpillar has the smallest, and MAN-B&W and Mitsubishi have similar ones.

Here is my table. I chose to include mean, median, standard deviation, and the IQR in my summary statistics.

```
dat.bal.cleaned |>
    group_by(Manufacturer) |>
    summarize(
        n = n(),
        Mean = mean(SPLog),
        SD = sd(SPLog),
        Median = median(SPLog),
        IQR = IQR(SPLog),
    )|>
    knitr::kable(digits = 2)
```

Table 2: Log of Ship Price by Manufacturer.

Manufacturer	n	Mean	SD	Median	IQR
Caterpillar	2	6.68	0.18	6.68	0.13
MAN-B&W	1588	7.15	0.26	7.16	0.36
Mitsubishi	63	7.03	0.23	7.04	0.34
Wartsila	166	7.03	0.32	6.99	0.55

From this table, we can see that despite having the same mean price, Mitsubishi has a higher median price than Wartsila. This means that Wartsila has some extreme outliers that drive up the mean price to make it higher than the median. MAN-B&W and Mitsubishi have almost the same IQR, despite Mitsubishi having a lower mean and median. Interstingly, Caterpillar only has two ships, which explains the relatively smaller IQR, and the mean and median being the exact same.

This table and graph confirm that the price of the ships does significantly change based on the manufacturer.

2.3.2 Part B

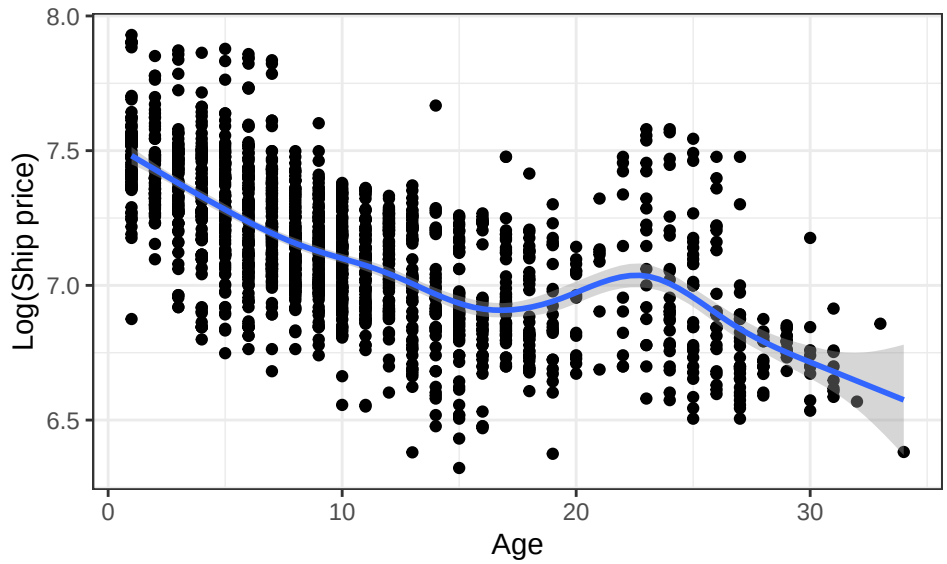
Choose a quantitative variable. Create a plot and corresponding numerical summary to assess whether or how ship prices vary with the selected quantitative variable.

Solution

I chose years as my quantitative variable. I made a scatter plot of years vs the log of ship price, and added a trend line to it.

```r
ggplot(data = dat.bal.cleaned) +
  geom_point(aes(x = Yas, y = SPLog)) +
  geom_smooth(aes(x = Yas, y = SPLog)) +
  theme_bw() +
  labs(x = "Age", y = "Log(Ship price)")
```

Figure 4: Log of ship price vs age of the ship



In this plot, it appears that the price of the ship decreases as it becomes older, which makes sense. There is a notable period around 21 years old where ship prices briefly increase before decreasing again. A possible reason for this is some larger and more expensive ships stopped being made, and thus they keep their price better than others.

```r
pearson <- cor(dat.bal.cleaned$SPLog, dat.bal.cleaned$Yas, method = "pearson")
kendall <- cor(dat.bal.cleaned$SPLog, dat.bal.cleaned$Yas, method = "kendall")
dat.bal.cleaned |>
    group_by(Yas) |>
    summarize(
        n = n(),
        Mean = mean(SPLog),
        SD = sd(SPLog),
        Median = median(SPLog),
        IQR = IQR(SPLog),
        Pearsons = (pearson),
        Kendalls = (kendall)
    )|>
    knitr::kable(digits = 2)
```

Table 3: Log of Ship Price by age of the ship.

Yas	n	Mean	SD	Median	IQR	Pearsons	Kendalls
1	101	7.49	0.16	7.48	0.16	-0.62	-0.49
2	67	7.43	0.16	7.40	0.17	-0.62	-0.49
3	90	7.38	0.19	7.41	0.21	-0.62	-0.49
4	100	7.30	0.21	7.32	0.25	-0.62	-0.49
5	102	7.29	0.21	7.28	0.30	-0.62	-0.49
6	123	7.25	0.24	7.26	0.32	-0.62	-0.49
7	125	7.21	0.19	7.19	0.24	-0.62	-0.49
8	124	7.15	0.15	7.14	0.20	-0.62	-0.49
9	119	7.12	0.16	7.13	0.22	-0.62	-0.49
10	147	7.09	0.16	7.10	0.22	-0.62	-0.49
11	90	7.06	0.17	7.08	0.25	-0.62	-0.49
12	90	7.08	0.16	7.10	0.28	-0.62	-0.49
13	64	7.03	0.22	7.03	0.35	-0.62	-0.49
14	45	6.93	0.23	6.91	0.20	-0.62	-0.49
15	63	6.91	0.22	6.90	0.28	-0.62	-0.49
16	41	6.90	0.23	6.88	0.41	-0.62	-0.49
17	55	6.95	0.17	6.93	0.19	-0.62	-0.49
18	43	6.90	0.19	6.82	0.33	-0.62	-0.49
19	23	6.90	0.23	6.86	0.39	-0.62	-0.49
20	14	6.92	0.17	6.96	0.29	-0.62	-0.49
21	7	7.05	0.20	7.11	0.13	-0.62	-0.49

Table 3: Log of Ship Price by age of the ship.

Yas	n	Mean	SD	Median	IQR	Pearsons	Kendalls
22	11	7.06	0.32	6.95	0.60	-0.62	-0.49
23	18	7.13	0.31	7.18	0.51	-0.62	-0.49
24	26	6.99	0.29	6.90	0.37	-0.62	-0.49
25	29	6.95	0.28	7.00	0.35	-0.62	-0.49
26	24	6.95	0.26	6.89	0.40	-0.62	-0.49
27	28	6.77	0.22	6.70	0.23	-0.62	-0.49
28	15	6.73	0.08	6.72	0.10	-0.62	-0.49
29	15	6.78	0.05	6.79	0.06	-0.62	-0.49
30	9	6.74	0.19	6.70	0.09	-0.62	-0.49
31	8	6.70	0.11	6.67	0.14	-0.62	-0.49
32	1	6.57	NA	6.57	0.00	-0.62	-0.49
33	1	6.86	NA	6.86	0.00	-0.62	-0.49
34	1	6.38	NA	6.38	0.00	-0.62	-0.49

Looking at the table, we can see that the average ship price decreases as the ship gets older. We notice that the average price of the ships bottoms out at 19 years, before increasing again until age 23. The average price goes below the 19 year old price again at age 28. We can see a large dropoff in the number of ships after they turn 11 years old. I had to calculate correlation outside of the summarize, because it was all NA values when the correlation was inside. Notably, the standard deviation of ships that are 32, 33, and 34 years old cannot be calculated. This is because there is only one ship in each of those ages.

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